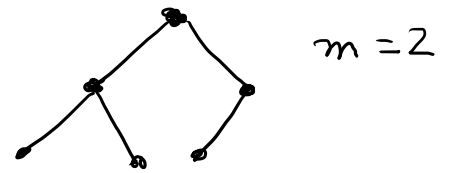
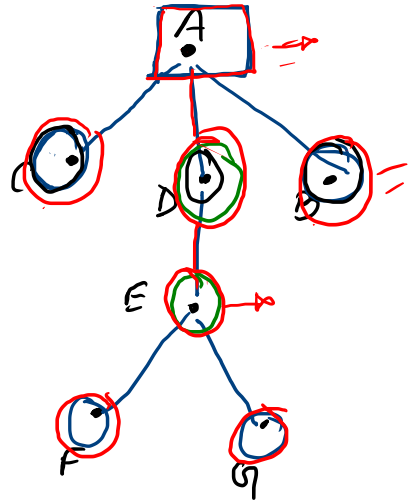
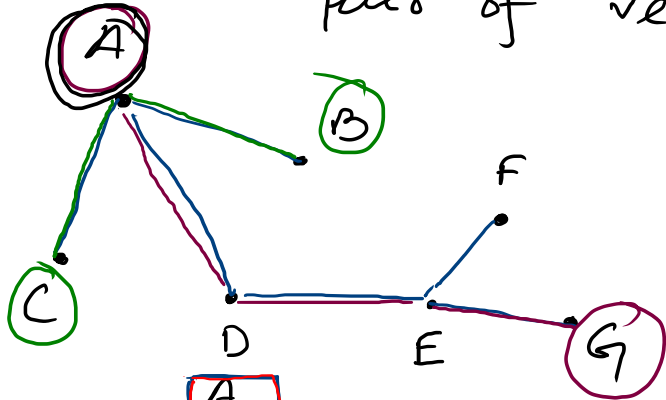


Ch-11.1 Trees.



Trees $\rightarrow$ CG $\rightarrow$ with no simple circuit
 $\rightarrow$ UG $\rightarrow$ is a tree iff there is a unique path b/w any pair of vertices.



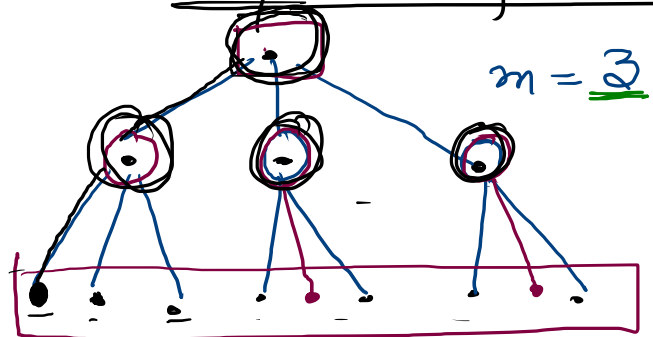
① Rooted tree $\rightarrow$ it has a node designated as the root / starting point and the edges are directed away from the root.

② Rooted m-ary tree

$\rightarrow$ the internal vertices will have no more than m -children.

$\rightarrow$ full m-ary tree $\rightarrow$ every internal vertex will have exactly m -children.

* Properties of trees.



* $T_n \rightarrow$ a tree with n vertices has $(n-1)$ edges.

* $T_m \rightarrow$ a full m -ary tree

- n vertices $\rightarrow i = \frac{(n-1)}{m}$ internal vertices.
 $\rightarrow l = \lceil \frac{(n-1)m+1}{m} \rceil$ leaves.

$$n = 13 \quad \boxed{h=2}$$

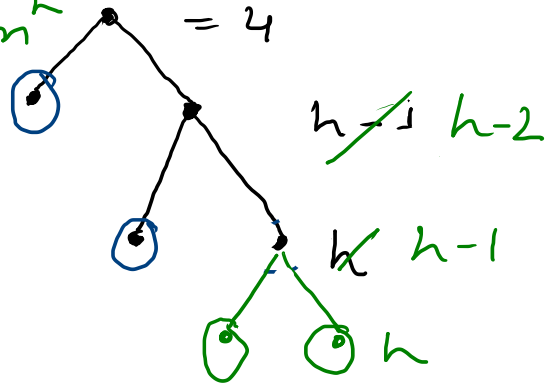
$$l = 12 = (n-1)$$

$$l = 9 \quad \left| \quad i = \frac{(n-1)}{m} \right.$$

$$= 3^2$$

$$= \frac{12}{3}$$

$$= 4$$



- i internal vertices $\rightarrow n = mi + 1$
 $\rightarrow l = (m-1)i + 1$

$\rightarrow n \rightarrow \frac{(ml-1)}{(m-1)}$
 $\rightarrow i \rightarrow \frac{(l-1)}{(m-1)}$

$$h = \lceil \log_m l \rceil$$

$$= \lceil \log_3 9 \rceil$$

$$= 2$$

⑩ level of a node $\rightarrow$ length of the path from root to the vertex/node

⑪ height of a tree $\rightarrow \forall n (\max(\text{level}_n))$

⑫ m-ary tree

$\hookrightarrow$ balanced? Unbalanced?

$\hookrightarrow$ all the leaves will be in the n or $(n-1)$ levels

